

EKmen exponentiell

C1

$$\begin{cases} u = u + iv \\ i\hbar(V - u_0) = \frac{d}{dz} \left(\psi(z) \frac{dV}{dz} \right) \quad (1) \\ \psi(z) = \psi_0 e^{-\kappa z} \end{cases}$$

Randwvler: $V(z=0) = 0, V(\infty) = u_0$

Steg 1: Setze $V = u(z) - u_0$

$$\Rightarrow \frac{d}{dz} (\psi_0 e^{-\kappa z} V') - i\hbar V = 0$$
$$\Leftrightarrow V'' - \kappa V' - \frac{i\hbar}{\psi_0} e^{\kappa z} V = 0$$

Steg 2: Introduce $t = \alpha e^{\kappa z/2}$

der $\alpha^2 = \frac{i\hbar}{\kappa^2 \psi_0}$

$$\Rightarrow t^2 \partial_t^2 V - t \partial_t V - t^2 V = 0 \quad (2)$$

Steg 3: Ansetze $V = tY$

$$\Rightarrow t^2 Y'' + tY' - (t+1)Y = 0$$

Modifizierte Bessel-Lsg. erdnung 1

$$\Rightarrow Y = c_1 I_1(t) + c_2 K_1(t)$$

$$\Rightarrow V = tY$$

Slag 4: Randvärder

När $z \rightarrow \infty$ för $t \rightarrow \infty$

$$\Rightarrow G=0$$

$$\Rightarrow V = C_2 e^{K_1(t)}$$

När $z=0$ för $t=\alpha$

$$\Rightarrow C_2 = \frac{U_0}{\alpha K_1(\alpha)}$$

Svar:
$$U(z) = U_0 \left[1 - \frac{t K_1(t)}{\alpha K_1(\alpha)} \right] \quad (3)$$

där $t = \alpha e^{Kz/2}$
 $\alpha = 2 \sqrt{\frac{1}{K^2 \gamma_0}}$

Notera: K_1 modifierad Bessel av andra slaget

Modifierad menas att vi använder
 $t = i t$.

För stora t $K_1(t) \sim \sqrt{\frac{\pi}{2t}} e^{-t}$

Alternativt Svar:
$$U(z) = U_0 \left[1 - \frac{e^{Kz/2} K_1(2\sqrt{1/\gamma_0} e^{Kz/2})}{K_1(2\sqrt{1/\gamma_0})} \right]$$

där $\rho = \frac{1}{K \alpha_0} = \frac{\sqrt{1/\gamma_0}}{K \alpha_0}$

(3)

Lösung: $U(z) = U_0 \left[1 - \frac{e^{Kz/2} \cdot K_1(\alpha e^{Kz/2})}{\alpha K_1(\alpha)} \right] \quad (3)$

der $\alpha = 2 \sqrt{\frac{if}{K^2 v_0}}$

Alternativ $\rho = \frac{1}{K v_0} = \frac{\sqrt{f}}{K \sqrt{v_0}}$
 $\Rightarrow \alpha = 2 \sqrt{if} \cdot \rho$

Markwinkel: $\Theta = \arg [Re(U'), Im(U')]$
 $= \tan^{-1} \left[\frac{Im(U')}{Re(U')} \right]$

$\frac{dU}{dz} = - \frac{U_0}{K_1(\alpha)} \frac{d}{dz} \left[e^{Kz/2} \cdot \overbrace{K_1(\alpha e^{Kz/2})}^{=t} \right]$

Var $\frac{d}{dz} (e^{Kz/2} K_1(t)) = \frac{K}{2} e^{Kz/2} K_1(t) + e^{Kz/2} \frac{d}{dz} K_1(t)$

ettersom $\frac{dt}{dz} = \frac{K}{2} t$

$\Rightarrow \frac{d}{dz} K_1(t) = \overbrace{K_1'(t)}^{=K_1'(t)} \cdot \frac{K}{2} t = \left[-K_0(t) - \frac{K_1(t)}{t} \right] \cdot \frac{K}{2} t$

$\Rightarrow \frac{dU}{dz} = \frac{U_0 K}{2 K_1(\alpha)} \alpha e^{Kz} K_0(\alpha e^{Kz/2})$

Markwinkel $\Rightarrow z \rightarrow 0$

$\Rightarrow \frac{dU}{dz} \Big|_0 = \frac{U_0 K}{2 K_1(\alpha)} \alpha K_0(\alpha)$

$\Rightarrow \frac{dU}{dz} \Big|_{z=0} = U_0 \sqrt{\frac{if}{v_0}} \cdot \frac{K_0(\alpha)}{K_1(\alpha)} \quad (4)$

Grens värde när $K \rightarrow \infty$.

$$\text{V: nr (4)} \quad \frac{dU}{dz}|_0 = U_0 \sqrt{i \frac{t}{v_0}} \frac{K_0(\alpha)}{K_1(\alpha)}, \quad \alpha = 2 \sqrt{i \frac{t}{K^2 v_0}}$$

$$\text{Om } K \rightarrow \infty \Rightarrow \alpha \rightarrow 0$$

$$\text{För } \alpha \rightarrow 0 \quad K_1(\alpha) \sim \frac{1}{\alpha} \\ K_0(\alpha) \sim -\ln(\alpha)$$

$$\text{Om } K \rightarrow \infty \Rightarrow \frac{dU}{dz}|_{z=0} \rightarrow U_0 \sqrt{i \frac{t}{v_0}} [-\alpha \ln(\alpha)] \quad \alpha = 2 \sqrt{i \frac{t}{K^2 v_0}} = \frac{C}{K} \\ \left(\begin{aligned} -\alpha \ln(\alpha) &= -\frac{C}{K} \ln\left(\frac{C}{K}\right) = -\frac{C}{K} [\ln C - \ln K] \\ &= \frac{C}{K} \ln K - \frac{C}{K} \ln C \sim \frac{C}{K} \ln K \end{aligned} \right) \quad C = 2 \sqrt{i \frac{t}{v_0}}$$

$$\Rightarrow \frac{dU}{dz}|_{z=0} \sim U_0 \sqrt{i \frac{t}{v_0}} \frac{C}{K} \ln K = U_0 \frac{2it}{v_0} \frac{C}{K} \ln K \\ \sim i \frac{\ln K}{K}$$

$$\text{i.e.} \quad \text{Im}(U'(z=0)) \rightarrow \frac{\ln K}{K} \quad \text{Om } K \rightarrow \infty \\ \text{Re}(U'(z=0)) \rightarrow 0 \quad \text{Om } K \rightarrow \infty$$

$$\Rightarrow \theta = \tan^{-1}\left(\frac{\text{Im}}{\text{Re}}\right) \rightarrow \tan^{-1}(\infty) \rightarrow \frac{\pi}{2}$$

$$\text{Närbere: } \frac{\text{Im}}{\text{Re}} \sim \ln K, \quad \theta = \tan^{-1}(\ln K)$$

$$\tan^{-1}(x) \approx \frac{\pi}{2} - \frac{1}{x} \quad \text{för stora } x$$

$$\Rightarrow \theta = \frac{\pi}{2} - \frac{1}{\ln K}$$

$$\Rightarrow \frac{\pi}{2} - \theta \sim \frac{1}{\ln K}$$

Vinkeln går mot 90°
Men översigt långsamt.

Vi har ett svar på formen

$$U(z) = U_0 \left[1 - \frac{e^{kz/2} K_1(2\sqrt{i} z) e^{kz/2} p}{K_1(2\sqrt{i} z) p} \right] \quad (3)$$

där $p = \frac{1}{k h_0} = \frac{1}{k \sqrt{h_0}}$

Men vi kan använda Kelvin funktionerna Ker & Kei

$$e^{-i\pi n/2} K_n(x e^{i\pi/4}) = Ker_n(x) + i Kei_n(x)$$

där vi vet att $\sqrt{i} = e^{i\pi/4}$.

$$\Rightarrow U(z) = U_0 \left[1 - e^{kz/2} \frac{Ker_1(\sqrt{z} p e^{kz/2}) + i Kei_1(-z p e^{kz/2})}{Ker_1(\sqrt{z} p) + i Kei_1(-z p)} \right] \quad (3b)$$

Undersök vinkeln genom $U'(z)$. Där vi vet

$$Ker_1(x) + i Kei_1(x) = -i K_1(x e^{i\pi/4})$$

$$\text{och } \frac{d}{dx} K_1(x) = -K_0(x) - \frac{1}{x} K_1(x)$$

$$\begin{aligned} \Rightarrow \frac{dU}{dz} &= - \frac{U_0 \frac{k}{2} e^{kz/2} i K_1(2\sqrt{i} z p e^{kz/2}) + U_0 e^{kz/2} \frac{dK_1(\dots)}{dz}}{(\dots)} \\ &= U_0 \frac{k}{2} e^{kz/2} 2 p e^{kz/2} \frac{i K_0(2 p e^{i\pi/4} e^{kz/2})}{i K_1(2 p)} \\ &= U_0 k p e^{kz/2} \sqrt{i} \frac{Ker_0(\sqrt{z} p e^{kz/2}) + i Kei_0(\sqrt{z} p e^{kz/2})}{Ker_1(\sqrt{z} p) + i Kei_1(\sqrt{z} p)} \end{aligned}$$

$$\Rightarrow \left. \frac{dU}{dz} \right|_{z=0} = U_0 k p \sqrt{i} \frac{Ker_0(\sqrt{z} p) + i Kei_0(\sqrt{z} p)}{Ker_1(\sqrt{z} p) + i Kei_1(\sqrt{z} p)}$$

alternativ form efter mycket algebra

$$\left. \frac{dU}{dz} \right|_{z=0} = \frac{U_0 k p}{\sqrt{z} (Ker_1(\dots) + i Kei_1(\dots))} \left[[Ker_0 Kei_0 + Kei_0 Ker_1 - Ker_0 Ker_1 + Ker_0 Kei_1] + i [Ker_0 Kei_0 + Kei_0 Ker_1 + Ker_0 Kei_1 + Kei_0 Ker_1 - Ker_0 Kei_1] \right]$$